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## **Diesel Engines V 4000 R 41**

### **Maintenance Schedule M050604/01E**

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Low operating and maintenance costs as well as operational reliability and availability depend on maintenance and servicing being carried out in compliance with our specifications and instructions.

The overall system, of which the engine is an integral part, must be maintained in such a way as to ensure trouble-free engine operation at all times. For this purpose always:

- ensure that sufficient fuel is available,
- ensure that the combustion air is dry and clean.

Our Product Support Service will always be available should assistance be required.

Preventive maintenance instructions.

- Special care should be taken to keep the machinery plant in a clean and serviceable condition at all times to facilitate early detection of possible leaks and prevent subsequent damage.
- Protect rubber and synthetic components from oil and fuel, never treat with organic detergents. Wipe with a dry cloth only.
- Always replace all seals and gaskets.

#### **MTU Maintenance Concept**

The maintenance concept for MTU products is a preventive maintenance concept.

Preventive maintenance facilitates advance planning and ensures a high degree of equipment availability.

The Maintenance Schedule is based on the power output defined in the individual orders, under consideration of the operating parameters established in the order's specifications.

The time intervals according to which the maintenance tasks and the relevant checks and tasks involved are to be completed are based on operational experience and can only serve as guidelines. Specific operational conditions may require modification of the Maintenance Schedule.

#### **Note:**

The engine manufacturer cannot be held responsible for the performance of the following maintenance tasks. The operator is responsible for carrying out the following maintenance tasks:

Fuel filter

The maintenance interval depends on the degree of fuel contamination. The fuel filter must be replaced after 2 years at the latest.

Battery

Battery maintenance depends on the degree of use and ambient temperature. The battery has to be checked according to the manufacturer's instructions, as it is not always a part of the MTU scope of supply.

Fluids and Lubricants:

Only those fluids and lubricants are to be used that are approved in the MTU Fluids and Lubricants Specification A001061. A special note must be made that, after maintenance or repair work on the cooling system, the operator must check the concentration of the corrosion inhibitor or the corrosion-inhibitor antifreeze.

#### **Major Overhaul**

Requires complete engine disassembly to facilitate repair/overhaul of all relevant components.

#### **Out-of-Service Periods**

If the engine is to remain out of service for several weeks, spray engine oil into the air-intake ducting. Actuate the engine shutdown system and bar the engine with the starting equipment when performing this task. Carry out maintenance tasks in accordance with the time limit schedule (see the Maintenance Frequency Chart). In addition, the engine must be barred manually every month. If the engine is to remain out of service for more than 1 month, carry out engine preservation in accordance MTU Fluids and Lubricants Specification, Publication No. A001061.

**Application group**

2A: Rail Traction

**Maintenance Frequency Chart**

|                   |                               |        |
|-------------------|-------------------------------|--------|
| Maintenance Level | W1: Operational checks, daily | X      |
|                   | W2: Operating hours           | 500    |
|                   | Limit, months                 | 6      |
|                   | W3: Operating hours           | 1 000  |
|                   | Limit, year                   | 1      |
|                   | W4: Operating hours           | 5 000  |
|                   | Limit, year(s)                | 2      |
|                   | W5: Operating hours           | 10 000 |
|                   | Limit, year(s)                | 6      |
|                   | W6: Operating hours           | 30 000 |
|                   | Limit, year(s)                | 18     |

**Maintenance Task Schedule**

|             | Maintenance Level W1<br>– Operational checks |                                                              |
|-------------|----------------------------------------------|--------------------------------------------------------------|
| G000.000.13 | Engine operation                             | Check speed (if gauges are provided)                         |
| G000.000.15 | Engine operation                             | Check pressures (if gauges are provided)                     |
| G000.000.17 | Engine operation                             | Check temperatures (if gauges are provided)                  |
| G000.000.19 | Engine operation                             | Check running noises                                         |
| G000.000.21 | Engine operation                             | Check visually for leaks and general condition of the engine |
| G083.101.03 | Fuel pre-filter                              | Drain water and contaminants                                 |
| G111.071.01 | Intercooler                                  | Check drain line for water discharge and obstructions        |
| G121.052.01 | Air filter                                   | Check service indicator                                      |
| G140.000.05 | Exhaust system                               | Check exhaust gas color                                      |
| G180.000.01 | Engine oil                                   | Check level                                                  |
| G202.000.03 | Engine coolant                               | Check level                                                  |
| G202.051.01 | Coolant pump                                 | Check relief bore for obstructions                           |
| G203.051.01 | Charge air coolant pump                      | Check relief bore for obstructions                           |

|                                                                                                                                               | Maintenance Levels                                          | W                                                                                                                                                                                                                                                                                                                                                         | 2 | 3 | 4 | 5 | 6 |
|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|
| G180.000.04                                                                                                                                   | Engine oil                                                  | Take sample and analyze                                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |
| G180.000.05                                                                                                                                   | Engine oil                                                  | Change <ul style="list-style-type: none"> <li>With oil category 1, with centrifugal oil filter<br/>Every 500 operating hours</li> <li>With oil category 2, with centrifugal oil filter<br/>Every 1000 operating hours</li> <li>With oil category 3, with centrifugal oil filter<br/>Every 1500 operating hours<br/>After 2 years at the latest</li> </ul> |   |   |   |   |   |
| G183.052.03                                                                                                                                   | Engine oil filter                                           | Replace <ul style="list-style-type: none"> <li>Accomplish when changing the oil</li> </ul>                                                                                                                                                                                                                                                                |   |   |   |   |   |
| G183.101.01                                                                                                                                   | Centrifugal oil filter                                      | Clean, check thickness of oil sludge layer<br>Replace paper sleeve                                                                                                                                                                                                                                                                                        |   |   |   |   |   |
| G055.050.01                                                                                                                                   | Valve gear                                                  | Check valve clearance                                                                                                                                                                                                                                                                                                                                     |   |   |   |   |   |
| G083.051.03                                                                                                                                   | Fuel duplex filter                                          | Change throw-away filter                                                                                                                                                                                                                                                                                                                                  |   |   |   |   |   |
| G083.101.09                                                                                                                                   | Fuel pre-filter                                             | Replace filter element and sealing ring                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |
| G121.051.03                                                                                                                                   | Air filter                                                  | Clean                                                                                                                                                                                                                                                                                                                                                     |   |   |   |   |   |
| ---                                                                                                                                           | Air ducting                                                 | Check ducting between air filter and turbocharger visually for leaks and damage                                                                                                                                                                                                                                                                           |   |   |   |   |   |
| G205.051.01                                                                                                                                   | Coolant cooler                                              | Check cooler elements for external contamination                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |
| G205.051.03                                                                                                                                   | Charge-air coolant cooler                                   | Check cooler elements for external contamination                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |
| G018.101.02                                                                                                                                   | Crankcase ventilation                                       | Replace filter elements                                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |
| G075.051.05                                                                                                                                   | Fuel injector(s)                                            | Replace                                                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |
| G121.051.01                                                                                                                                   | Air filter                                                  | Replace filter elements                                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |
| G175.100.01                                                                                                                                   | Flame-torch starting system                                 | Replace flame torch (if provided)                                                                                                                                                                                                                                                                                                                         |   |   |   |   |   |
| G202.000.05                                                                                                                                   | Engine coolant                                              | With water-soluble corrosion inhibitor:<br>Take sample and analyze <ul style="list-style-type: none"> <li>Change coolant, see MTU Fluids and Lubricants Specification</li> </ul>                                                                                                                                                                          |   |   |   |   |   |
| G213.051.01                                                                                                                                   | Charging generator                                          | Check condition of coupling, replace if necessary                                                                                                                                                                                                                                                                                                         |   |   |   |   |   |
| G223.052.01                                                                                                                                   | Hydraulic pump drive(s)<br>(Equipment carrier / Crankshaft) | Check condition of coupling(s), replace if necessary                                                                                                                                                                                                                                                                                                      |   |   |   |   |   |
| G252.051.03                                                                                                                                   | Coupling (Küsel)                                            | Check radial clearance                                                                                                                                                                                                                                                                                                                                    |   |   |   |   |   |
| ---                                                                                                                                           | Electrical preheating unit                                  | Check pump for leaks, (Sichtkontrolle)<br>replace rotary seal if necessary, engine Motor (if provided)                                                                                                                                                                                                                                                    |   |   |   |   |   |
| <b>Prior to commencing W5 maintenance, run the engine and record all measured values, then drain engine coolant and flush cooling system!</b> |                                                             |                                                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |
|                                                                                                                                               | Valve gear                                                  | Check rocker arms, valve bridges, swing followers and swing follower bearings for wear                                                                                                                                                                                                                                                                    |   |   |   |   |   |
|                                                                                                                                               | Cylinder heads                                              | Replace                                                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |
|                                                                                                                                               | Running gear                                                | Check piston crowns (visual inspection)                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |
|                                                                                                                                               | Crankcase                                                   | Check cylinder liner wear pattern                                                                                                                                                                                                                                                                                                                         |   |   |   |   |   |
|                                                                                                                                               | Vibration damper                                            | Replace                                                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |
|                                                                                                                                               | Turbocharger                                                | Replace                                                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |
|                                                                                                                                               | Air ducting                                                 | Clean                                                                                                                                                                                                                                                                                                                                                     |   |   |   |   |   |
|                                                                                                                                               | Intercooler                                                 | Replace                                                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |
|                                                                                                                                               | Starter                                                     | Replace                                                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |
|                                                                                                                                               | Charging generator                                          | Replace                                                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |
|                                                                                                                                               | HP fuel delivery pump                                       | Replace                                                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |
|                                                                                                                                               | HP fuel delivery-pump sensor                                | Replace                                                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |
|                                                                                                                                               | Engine coolant                                              | With antifreeze:<br>Take sample and analyze <ul style="list-style-type: none"> <li>Change coolant, see MTU Fluids and Lubricants Specification</li> </ul>                                                                                                                                                                                                 |   |   |   |   |   |
|                                                                                                                                               | Coolant pump                                                | Replace                                                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |
|                                                                                                                                               | Charge air coolant pump                                     | Replace                                                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |
|                                                                                                                                               | Coolant cooler                                              | Clean and leak test                                                                                                                                                                                                                                                                                                                                       |   |   |   |   |   |
|                                                                                                                                               | Charge-air coolant cooler                                   | Clean and leak test                                                                                                                                                                                                                                                                                                                                       |   |   |   |   |   |
|                                                                                                                                               | Electrical preheating unit                                  | Maintenance is the same as for the engine (if provided)<br>Carry out visual and acoustic check of the three-phase motor, replace smooth bearing(s) if necessary<br>Re-grease motor's smooth bearing(s)                                                                                                                                                    |   |   |   |   |   |
|                                                                                                                                               | Engine oil heat exchanger                                   | Remove, clean, leak test and replace seals                                                                                                                                                                                                                                                                                                                |   |   |   |   |   |
|                                                                                                                                               | Engine mounts                                               | Check set adjustment of the central buffer                                                                                                                                                                                                                                                                                                                |   |   |   |   |   |

|  |                                                                |                                                                                                                                                                                              |               |
|--|----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
|  | Hydraulic pump drive(s)<br>(Equipment carrier /<br>Crankshaft) | Replace coupling(s)                                                                                                                                                                          |               |
|  | Electrical fuel delivery pump                                  | Pump operates for a maximum of 1% of the engine operating time<br>D.C. motor<br>– Check length of carbon brushes and condition of commutator<br>– Remove carbon residue from engine interior | (if provided) |
|  | Centrifugal oil filter                                         | Check, replace bearings if necessary                                                                                                                                                         |               |
|  | Coupling (Küsel)                                               | Overhaul                                                                                                                                                                                     |               |
|  | Major Overhaul                                                 | Total disassembly of the engine; inspection / repair of all components                                                                                                                       |               |